

Matthew Minwoo Kim

✉ matthewkim.stu@gmail.com 🌐 matmkim.me

Education

Seoul National University <i>B.S. in Computer Science and Engineering</i> - GPA: 4.21/4.3 (76 credits)	Mar. 2023 – Present Seoul, South Korea
Korea Science Academy of KAIST <i>High School Diploma, Computer Science</i> - GPA: 4.14/4.3 (top 10% of the class)	Mar. 2020 – Feb. 2023 Busan, South Korea
University of Cambridge <i>International Summer School - Business and Entrepreneurship</i>	Summer 2022 Cambridge, United Kingdom

Research Experience

Intelligent Data Systems Lab , Seoul National University <i>Undergraduate Research Opportunity Program (Advisor: Prof. Sang-goo Lee)</i> ○ Researched with topic of "Text-to-SQL Generation using Intermediate Representation" and "Resolving Ambiguity in Text-to-SQL Generation" (Manuscript in preparation)	Jul. 2024 – Jul. 2025
Computer Science Research Group , Korea Science Academy of KAIST <i>Student Researcher</i> ○ Researched with the topic of "A Secret Sharing Scheme without Third-Party Dealers" ○ Resulted in Regeneron ISEF South Korea Finalist	Feb. 2021 – Feb. 2022

Professional Experience

Republic of Korea Air Force <i>AI Researcher</i> ○ Leading R&D of military-specific computer vision and STT systems	Oct. 2024 – Jul. 2026
AIBIZON AI Lab <i>Research Team Intern</i> ○ Conducted an empirical study on RAPTOR (Recursive Abstractive Processing for Tree-Organized Retrieval) ○ Developed CRDT-based sync program between Slack and a database	Mar. 2024 – Apr. 2024
Hyperithm <i>Research Team & Infra Team Intern</i> ○ Developed HFT-optimized exchange data feed and trading strategy tools using Rust and Python	Jun. 2023 – Aug. 2023

Awards & Achievements

Army Chief of Staff Award , Military AI Contest (MAICON) ○ 2nd place, national-level AI competition	2025
South Korea Finalist , Regeneron ISEF (International Science and Engineering Fair) ○ Systems Software category (Team Leader)	2022
Grand Prize , R&E Project Presentation Contest for the Scientific Talent ○ Ministry of Science and ICT Award	2021

Publications

- [1] Seokhun Ji, Euisuh Jeong, **Minwoo Kim**, Hyojin Lee, "Deep Learning for Pavement Management System: Proposing an Automated Pipeline for Pavement Condition Index (PCI) Assessment", in *Journal of the Korean Society of Airport*, 1(1), 52-65.
- [2] Euisuh Jeong, Seokhun Ji, **Minwoo Kim**, Hyojin Lee, "ROKAF Runway Crack Dataset: Construction and Application of a Large-Scale AI-Based Runway Defect Detection Dataset", in *Journal of the Korean Society of Airport*, (Accepted, 2025)

Certificates

SQL Developer	Jun 2024
Craftsman Information Processing	Apr 2024

Skills

- **Languages:** Korean (Native), English (Advanced)
- **Programming Languages:** Python(PyTorch, TensorFlow), C++, Java, Typescript, Rust

Research Interests

- Computer Vision
- Multimodal
- NLP
- Algebra